Enhancing Heuristic and Metaheuristic Solutions for the Weapon-Target Assignment Problem via Iterative Refinement and Hybrid Genetic Algorithms

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***Abstract*—The Weapon-Target Assignment (WTA) problem asks how to allocate a finite set of weapons to a set of targets so as to minimise the expected total survival value of those targets. It is NP-complete, and practical instances arising in air defence and ground-combat scenarios can have hundreds of weapons and targets, rendering exact methods infeasible within real-time constraints. This paper presents two algorithm engineering contributions. First, we augment the classic Maximum Marginal Return (MMR) greedy heuristic with (i) tie-breaking by target value, (ii) a 1-opt iterative local-search pass, and (iii) a 2-opt swap-search pass, yielding the MMR-IR algorithm. Second, we redesign the standard Genetic Algorithm (GA) with (i) MMR-seeded initialisation, (ii) elitism with stagnation-triggered re-diversification, and (iii) threat-proportional mutation, yielding the Hybrid GA. Both modifications are evaluated on a structured synthetic benchmark of 180 instances spanning nine categories (three size tiers $\times$ three weapon-availability scenarios). MMR-IR improves MMR in all nine categories, with per-category gains ranging from 0.11% to 3.15%. On balanced instances, MMR-IR improves MMR by 2.44%, 0.46%, and 0.13% in small, medium, and large tiers, respectively. Hybrid GA changes GA performance by -2.04%, 28.85%, and 37.70% in the same tiers, showing strong gains at medium/large scale but mixed small-tier behaviour. Using Wilcoxon signed-rank tests ($\alpha = 0.05$, alternative = greater), MMR-IR is significant in $9/9$ categories and Hybrid GA in $6/9$ categories (all three small-tier categories are non-significant for Hybrid GA).**

I. Introduction

The Weapon-Target Assignment (WTA) problem is a central combinatorial optimisation problem in military operations research. Given a set of weapons and a set of incoming targets, each weapon-target pair is characterised by a kill probability p_{ij}, and each target carries a strategic value V_j. The objective is to find an integer allocation x_{ij} (number of weapons of type i assigned to target j) that minimises the expected residual threat $\sum_j V_j \prod_i (1-p_{ij})^{x_{ij}}$, subject to weapons not exceeding their available stock.

Real-world systems such as Israel's Iron Dome require near-optimal WTA decisions within seconds: incoming rockets must be assigned to interceptors before they reach their targets [1]. This tension between solution quality and runtime is the core algorithmic challenge.

WTA belongs to the class of NP-complete problems [2], [3], so practical solvers rely on heuristics and metaheuristics. Two widely studied baseline approaches are the Maximum Marginal Return (MMR) greedy heuristic and a Genetic Algorithm (GA) [1], [2]. MMR is extremely fast but commits irrevocably to early greedy choices. Standard GA explores the solution space more broadly but suffers from slow convergence, solution loss between generations, and uninformed mutation in large-dimensional instances.

This paper makes the following contributions:

- We propose **MMR-IR**, an extension of MMR that adds tie-aware greedy allocation followed by 1-opt and 2-opt local search, with $O(1)$ incremental objective evaluation per trial.
- We propose **Hybrid GA**, which addresses three identified weaknesses of standard GA through MMR-seeded initialisation, elitism with stagnation detection, and a novel threat-proportional mutation operator.
- We design and open-source a structured 180-instance synthetic benchmark covering small, medium, and large problem sizes under balanced, scarce, and weapon-rich resource scenarios.
- We provide a comprehensive experimental evaluation including one-sided Wilcoxon signed-rank testing; MMR-IR is significant in $9/9$ categories, while Hybrid GA is significant in $6/9$ and shows its strongest gains on medium and large tiers.

The remainder of the paper is organised as follows. Section II formalises the WTA problem and proves its NP-completeness. Section III surveys existing solution approaches. Sections IV-A and IV-B describe the baseline algorithms in detail. Section V presents our proposed modifications. Section VI describes the experimental setup. Section VII presents and discusses the results. Section VIII concludes.

II. Problem Formulation

A. Parameters and Decision Variables

Let $W = \{1, \ldots, m\}$ denote the set of weapon types and $T = \{1, \ldots, n\}$ the set of targets. Each target $j \in T$ has a strategic military value $V_j > 0$. For each weapon-target pair (i, j), let $p_{ij} \in [0, 1]$ denote the single-shot kill probability of weapon type i against target j, and let w_i denote the total stock of weapon type i. The decision variable $x_{ij} \in \mathbb{Z}^{\geq 0}$ represents the number of weapons of type i assigned to target j.

B. Objective Function

Under the standard independence assumption, the probability that target j survives all assigned weapons is $\prod_{i=1}^{m}(1 - p_{ij})^{x_{ij}}$. The static WTA problem (target-based formulation) is [2], [4]:

$$\text{Minimise} \quad Z = \sum_{j=1}^{n} V_j \prod_{i=1}^{m}(1 - p_{ij})^{x_{ij}} \tag{1}$$

$$\text{subject to} \quad \sum_{j=1}^{n} x_{ij} \leq w_i, \quad \forall\, i \in W \tag{2}$$

$$x_{ij} \in \mathbb{Z}^{\geq 0}, \quad \forall\, i \in W,\ j \in T \tag{3}$$

Equivalently, one may maximise total expected damage $\sum_j V_j \left[1 - \prod_i (1 - p_{ij})^{x_{ij}}\right]$; both formulations are equivalent.

C. NP-Completeness of WTA

We establish NP-completeness via two polynomial-time reductions. The primary formal citation is the classical conference result [3]; we also reviewed a worked pedagogical subset-product construction in [5].

1) Reduction 1: Subset Product $\leq_p$ WTA: The Subset Product problem — given a set $A = \{a_1, \ldots, a_m\}$ and a target b, decide whether a subset $X \subseteq A$ with $\prod_{x \in X} a_x = b$ exists — is NP-complete. Let $P = \prod_{i=1}^{m} a_i$. Construct a WTA instance with two targets ($V_1 = V_2 = P$) and $m + 2$ weapons with survival probabilities $q_{i,1} = q_{i,2} = 1/a_i$ for $i \leq m$, $q_{\alpha,1} = q_{\alpha,2} = b/(2P)$, and $q_{\beta,1} = q_{\beta,2} = 1/(2b)$. Set threshold $k = 1$.

Forward direction. If $X \subseteq \{1, \ldots, m\}$ satisfies $\prod_{x \in X} a_x = b$, assign weapons in X and W_α to T_1, and the rest to T_2. Then $D = P \cdot \frac{b}{2P} \cdot \frac{1}{b} + P \cdot \frac{1}{2b} \cdot \frac{b}{P} = \frac{1}{2} + \frac{1}{2} = 1 \leq k$. ✓

Backward direction. Suppose a WTA solution achieves $D \leq 1$. One shows that W_α and W_β must target different targets (assigning both to the same target yields $D > 1$, a contradiction). With $W_\alpha \to T_1$ and $W_\beta \to T_2$, letting $z = \prod_{x \in X} a_x$:

$$\frac{b}{2z} + \frac{z}{2b} \leq 1 \implies (b - z)^2 \leq 0 \implies z = b,$$

so X is a valid solution to Subset Product. ✓

The construction takes $O(m)$ time, so the reduction is polynomial.

2) Reduction 2: WTA $\leq_p$ Multiple Knapsack: Given a WTA instance with m targets and n weapons, define m knapsacks (one per target) with items corresponding to weapons. Set the profit of placing weapon i in knapsack j to $v_{ij} = -\log(1 - p_{ij})$, and the weight to 0 (or 1 to encode the weapon stock constraint w_i by replicating item i exactly w_i times).

Minimising $\prod q_{ij}^{x_{ij}}$ over j is equivalent to maximising $\sum_i x_{ij} v_{ij}$ (after taking logarithms), which is exactly the Multiple Knapsack objective. The transformation takes $O(nm)$ time.

Together, these two reductions show that WTA is NP-hard and lies in NP (a solution can be verified in polynomial time); hence **WTA is NP-complete**.

III. Survey of Existing Algorithms

Algorithms for WTA fall into five broad categories:

Exact algorithms guarantee globally optimal solutions but have exponential worst-case complexity. Branch-and-bound with LP/IP relaxation [2] runs in $O(2^{mn})$ in the worst case. Branch-price-and-cut with column generation [6] scales better for structured instances. Branch-and-adjust uses a compact linear under-approximation [7].

Approximation algorithms provide provable bounds in polynomial time. Piecewise linear approximation of the survival function achieves $O(mnB)$ complexity [7], [8]. LP relaxation followed by rounding yields polynomial complexity but no tight approximation ratio is known. Lagrangian relaxation decomposes the problem target-by-target and iterates via subgradient optimisation [9], [10].

Randomised algorithms trade quality for generality. Monte Carlo sampling evaluates random assignments across S trials at cost $O(Smn)$ [1]. Randomised marginal-return heuristics apply probabilistic selection to MMR [11].

Heuristic algorithms are fast and problem-specific. MMR (Maximum Marginal Return) [2], [12] runs in $O(W^2T)$. Very Large-Scale Neighbourhood (VLSN) search constructs improvement graphs and applies multi-exchange moves in $O(m^2n^2)$ [2]. MRBCH [12] extends MMR to sensor-weapon-target triplets.

Metaheuristic algorithms balance exploration and exploitation for large instances. Genetic algorithms [1], [13] use crossover and mutation over a population. Ant Colony Optimisation (ACO) [14]–[16] uses pheromone trails for constructive search. Multi-objective variants handle Pareto-optimal WTA fronts [17].

Table I summarises the complexity and applicability of each category.

TABLE I: Algorithm Category Comparison for WTA

Category	Complexity	Max Scale	Use Case
Exact	$O(2^{mn})$	< 1000 vars	Global optimality
Approximation	$O(\text{poly})$	> 10000	Theoretical bounds
Randomised	$O(S \cdot mn)$	Any	Baseline
Heuristic	$O(mn)$–$O(m^2n^2)$	> 1000	Real-time
Metaheuristic	$O(I \cdot P \cdot mn)$	> 10000	Large-scale

IV. Baseline Algorithms

A. Maximum Marginal Return (MMR)

MMR is a greedy heuristic that iteratively assigns the unallocated weapon with the maximum expected reduction in the total survival value [2]. At each step it scans all unallocated weapons and all targets, computing the *marginal return*:

$$\text{decrease}(w,t) = V_t^{\text{current}} \times p_{wt},$$

selects the (w^*, t^*) pair with the largest decrease, assigns w^* to t^*, and updates $V_{t^*}^{\text{current}} \leftarrow V_{t^*}^{\text{current}} - \text{decrease}(w^*, t^*)$. This process repeats until all weapons are allocated.

Algorithm 1 gives the full pseudocode.

Algorithm 1 MMR Greedy Heuristic

```
V^cur[t] ← V[t] for all t; pool ← all weapons
while pool is non-empty do
    (w*, t*) ← arg max_{w∈pool, t} V^cur[t] × p_wt
    x[w*] ← t*;   V^cur[t*] ← V^cur[t*] − V^cur[t*] × p_{w*,t*}
    Remove w* from pool
end while
return x
```

Complexity. $O(W^2T)$ where W is the number of weapons and T the number of targets. MMR is deterministic and extremely fast but can commit to sub-optimal early decisions that cannot be corrected.

Illustrative Example: Consider a 3×3 instance with target values $V = (10, 20, 30)$ and kill probabilities:

$$P = \begin{pmatrix} 0.3 & 0.2 & 0.1 \\ 0.1 & 0.4 & 0.2 \\ 0.2 & 0.3 & 0.5 \end{pmatrix}$$

(rows = weapons W0, W1, W2; columns = targets T0, T1, T2).

Round 1. All nine decreases are computed; W2→T2 gives the maximum $30 \times 0.5 = 15.0$. Assign W2→T2; $V[\text{T2}] \leftarrow 15$.

Round 2. W1→T1 gives the maximum $20 \times 0.4 = 8.0$. Assign W1→T1; $V[\text{T1}] \leftarrow 12$.

Round 3. W0→T0 gives the maximum $10 \times 0.3 = 3.0$. Final allocation: $[0, 1, 2]$ (W0→T0, W1→T1, W2→T2).

B. Standard Genetic Algorithm (GA)

The standard GA encodes a solution as a chromosome of length W, where chromosome$[i] \in \{0, \ldots, T-1\}$ is the target assigned to weapon i [1], [13]. The population size is set to $\max(W, T)$. The GA runs for a fixed time budget rather than a fixed number of generations.

Selection. Tournament selection with tournament size $k = 3$: randomly sample k individuals, return the one with lowest fitness.

Crossover. Uniform crossover with probability $P_c = 0.8$: each gene of an offspring is independently drawn from one of the two parents with equal probability.

Mutation. Each gene mutates with probability $P_m = 1/W$: replace the assigned target with a uniformly random target.

Fitness. Lower is better; fitness equals the WTA objective value Z (Eq. 1) evaluated for the chromosome.

Algorithm 2 gives the pseudocode.

Algorithm 2 Standard Genetic Algorithm for WTA

```
pop ← RandomPopulation(max(W, T))
best ← arg min{Z(c) : c ∈ pop}
while time budget not exhausted do
    p1, p2 ← TournamentSelect(pop)
    c1, c2 ← UniformCrossover(p1, p2)
    c1, c2 ← Mutate(c1), Mutate(c2)
    if Z(c1) < Z(best) then best ← c1
    end if
    if Z(c2) < Z(best) then best ← c2
    end if
    Replace population with offspring
end while
return best
```

V. Proposed Modifications

A. MMR-IR: MMR with Iterative Refinement

The greedy MMR solution is a good starting point but is suboptimal because each weapon is assigned exactly once and early commitments cannot be revised. MMR-IR wraps MMR with two post-processing local-search phases.

1) Enhancement 1: Tie-Breaking by Target Value: When two weapon-target pairs (w_1, t_1) and (w_2, t_2) have equal marginal return, MMR-IR prefers the pair whose target has the higher current value $V^{\text{cur}}[t]$. This ensures that coverage of high-value targets is never sacrificed to a tie-breaking tie.

2) Enhancement 2: 1-opt Local Search: After MMR greedy terminates, MMR-IR enters a 1-opt phase. In each pass, every weapon w is considered for reassignment to every target $t \neq x[w]$. If the resulting objective Z' is strictly less than the current Z, the reassignment is accepted. Passes repeat until no improvement is found, with a maximum of 20 passes to bound runtime.

Incremental evaluation. Maintaining an array $\text{surv}[j] = \prod_{i:\, x[i]=j}(1 - p_{ij})$ allows δZ computation in $O(1)$ per trial instead of $O(WT)$:

$$\delta Z = V_{t'}\big[\text{surv}[t'] \cdot (1 - p_{wt'}) - \text{surv}[t']\big] + V_t\big[\text{surv}[t]/(1 - p_{wt}) - \text{surv}[t]\big],$$

where $t = x[w]$ (current target) and t' (candidate target).

3) Enhancement 3: 2-opt Swap Search: After 1-opt converges, a 2-opt phase tries *pairwise swaps*: select two weapons w_a, w_b with $x[w_a] = t_a$, $x[w_b] = t_b$, and test swapping them so that $w_a \to t_b$ and $w_b \to t_a$. Swapping can escape 1-opt local optima that single reassignments cannot: moving w_a alone may worsen the objective on t_a, and moving w_b alone may worsen it on t_b, but swapping both simultaneously may improve the total. A maximum of 3 passes is applied, with $O(1)$ delta evaluation per candidate.

B. Hybrid GA

Standard GA on large WTA instances (e.g., $W = 375$, $T = 250$) suffers from three compounding weaknesses: (1) *Random initialisation*: in a space of T^W chromosomes, the initial population is far from any good solution; (2) *No elitism*: the best known solution can be overwritten when the population is replaced by offspring; (3) *Blind mutation*: uniformly random target reassignment ignores the relative threat levels of individual targets.

We address each weakness with a targeted modification.

1) Fix 1: MMR-Seeded Initialisation: The first individual in the initial population is set to the MMR greedy solution, which is near-optimal in $O(W^2T)$ time. The remaining individuals are initialised randomly to preserve population diversity. For large instances, this warm-start reduces wasted early generations spent escaping random-noise starting points.

2) Fix 2: Elitism with Stagnation Detection: At each generation replacement, the top 10% of the current population (by fitness) is carried forward unchanged into the next population. In addition, a stagnation counter tracks consecutive generations with no improvement in the best-known fitness. If the counter reaches 5, a fraction of the population (excluding elites) is replaced with freshly randomised individuals to reintroduce diversity. This prevents premature convergence while still preserving the best solutions found.

3) Fix 3: Threat-Proportional Mutation (Novel WTA-Specific Operator): Instead of selecting a mutation target uniformly at random, we bias the selection proportionally to the *remaining threat* of each target:

$$\text{threat}[j] = V_j \prod_{i:\, x[i]=j} (1 - p_{ij}).$$

The mutation target is drawn from the categorical distribution $P(\text{mutate to } j) \propto \text{threat}[j]$. Intuitively, targets with high remaining survival value (high V_j and few or ineffective weapons assigned) attract more mutations than already well-covered targets. This is a novel, WTA-specific contribution: it embeds domain knowledge directly into the search operator.

Concrete example. Suppose T1 has $V_1 = 100$ and current survival 0.6 ($\text{threat}_1 = 60$), while T2 has $V_2 = 20$ and survival 0.1 ($\text{threat}_2 = 2$). T1 is $30\times$ more likely to receive a mutated weapon than T2.

VI. Experimental Setup

A. Dataset Design

We constructed a fully synthetic benchmark to enable controlled scaling analysis across two independent dimensions: *problem size* (small, medium, large) and *weapon availability* (balanced, scarce, rich). Existing WTA benchmark datasets are scarce in the literature and rarely cover all three size tiers simultaneously; our design therefore deliberately fills this gap.

Table II specifies all nine categories.

Instance parameters are drawn from fixed distributions: target values $V_j \sim \mathcal{U}(10, 100)$ (integer-valued), kill probabilities $p_{ij} \sim \mathcal{U}(0.05, 0.90)$. All 180 instances are generated with a fixed random seed (42) for full reproducibility.

TABLE II: Benchmark Dataset: 9 Categories $\times$ 20 Instances = 180 Total

Tier	Scenario	W	T	Instances
Small	Balanced ($W = T$)	20	20	20
Small	Scarce ($W < T$)	10	20	20
Small	Rich ($W > T$)	30	20	20
Medium	Balanced	100	100	20
Medium	Scarce	50	100	20
Medium	Rich	150	100	20
Large	Balanced	250	250	20
Large	Scarce	125	250	20
Large	Rich	375	250	20
Total				180

Scenario rationale. The *balanced* scenario ($W = T$) represents the standard benchmark where each weapon covers one target on average. The *scarce* scenario ($W < T$) is the hardest: with fewer weapons than targets, some targets receive no coverage and survive with probability 1. The *rich* scenario ($W > T$) is the easiest: surplus weapons allow thorough multi-hit coverage, driving residual threat lower.

B. Algorithms and Settings

Four algorithms are evaluated on every instance:

1) **MMR (original)**: Maximum Marginal Return greedy heuristic, faithful reimplementation following Ahuja et al. [2] and the algorithm description in [4].
2) **MMR-IR (modified)**: MMR with tie-breaking, 1-opt (≤ 20 passes), and 2-opt (≤ 3 passes) as described in Section V-A.
3) **GA (original)**: Standard GA with tournament selection ($k = 3$), uniform crossover ($P_c = 0.8$), uniform mutation ($P_m = 1/W$), and time-budget termination.
4) **Hybrid GA (modified)**: GA with MMR seed, 10% elitism, stagnation detection (threshold = 5 generations), and threat-proportional mutation; same time budget as GA.

The time budget is set proportionally to problem size. The performance metric is the objective value Z (lower is better). All algorithms are implemented in Python with a shared random seed (42) for stochastic methods.

C. Statistical Validation

With 20 instances per category, we apply the **Wilcoxon signed-rank test** using the one-sided alternative "original $>$ modified" ($\alpha = 0.05$), matching the analysis pipeline in `analysis.py`. Tests are run category-wise for both pairs (MMR vs. MMR-IR and GA vs. Hybrid GA). The Wilcoxon test is appropriate here because it makes no normality assumption and is robust to non-Gaussian objective distributions.

VII. Experimental Results

Four algorithms produce a total of $180 \times 4 = 720$ result rows. We report core comparative findings first, then additional diagnostics.

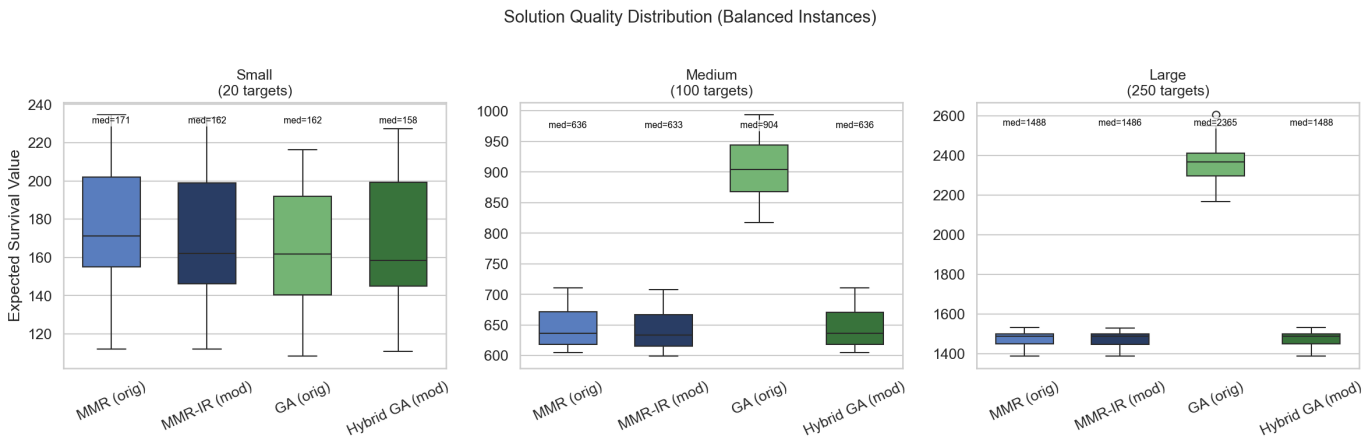


Fig. 1: Distribution of objective values (Z, lower is better) for all four algorithms across small, medium, and large tiers (balanced scenario). Boxes show IQR; whiskers extend to 1.5×IQR; circles are outliers.

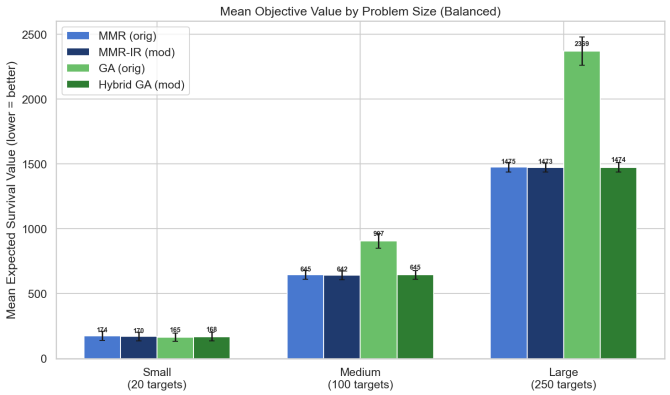


Fig. 2: Mean expected survival value per algorithm per balanced-tier benchmark (lower is better).

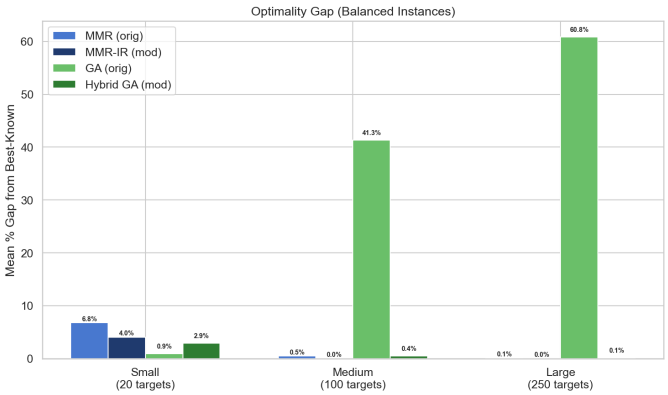


Fig. 3: Relative gap from best-known solution, where best-known is the minimum value found across all four algorithms for each instance.

A. Solution Quality Distribution

Figure 1 shows box plots of objective values by algorithm and tier. Three observations stand out:

- Dispersion increases with tier size, confirming that large instances are significantly harder than small ones.
- The MMR-IR median is consistently below MMR across all tiers, demonstrating stable quality gain from the local-search post-processing.
- GA (original) exhibits high spread on medium and large tiers; Hybrid GA produces both lower medians and tighter distributions, reflecting the stabilising effect of elitism and seeded initialisation.

Across all nine categories (three tiers × three scenarios), the same distribution pattern persists. For GA variants, separation is strongest in medium and large tiers, while small-tier overlap is visibly larger.

B. Mean Objective Value and Optimality Gap

Figure 2 shows that the quality gap between GA (original) and Hybrid GA widens as instance size grows. In the balanced small tier, GA (original) has lower mean Z than Hybrid GA (164.70 vs. 168.00), but this relation reverses strongly in medium and large tiers (906.80 vs. 644.51, and 2369.05 vs. 1474.28). Figure 3 confirms that GA (original) has the largest gap from best-known solutions on large-tier instances.

C. Improvement Percentages

Table III summarises balanced-scenario mean improvement of each modified algorithm over its baseline.

TABLE III: Balanced-Scenario Mean Improvement (%) of Modified Algorithms

	Mean Improvement (%)		
Pair	Small	Medium	Large
MMR-IR vs. MMR	2.44	0.46	0.13
Hybrid GA vs. GA	-2.04	28.85	37.70

Figure 4 confirms that MMR-IR remains positive in every balanced tier, while Hybrid GA is mixed in small tier and

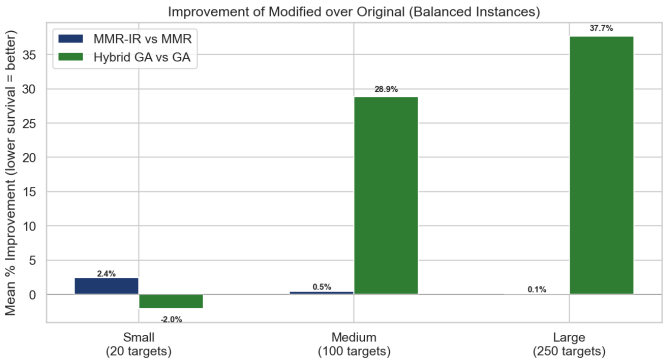


Fig. 4: Percentage improvement of modified algorithms over their baselines, broken down by problem size tier (balanced scenario).

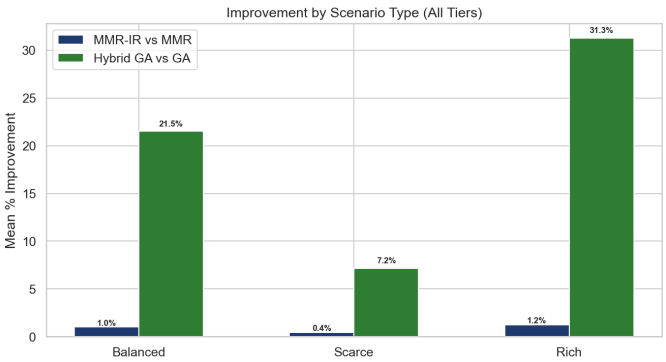


Fig. 5: Percentage improvement of modified algorithms over their baselines, broken down by scenario type (balanced / scarce / rich), averaged across tiers.

strongly positive in medium and large tiers. Scenario-wise, Hybrid GA is positive in all medium/large categories and mixed in small categories (small_balanced and small_scarce are negative, small_rich is positive). MMR-IR remains positive in all nine categories.

D. Runtime Scalability

Figure 6 shows that on a log-log scale, MMR and MMR-IR remain orders of magnitude faster than GA variants. GA variants approach their time budget cap as size grows; Hybrid GA still achieves superior quality within that same budget due to the warm start and directed search operators. The log-log slopes confirm that both MMR and MMR-IR scale sub-quadratically, while GA variants are dominated by the fixed time budget.

E. Additional Results: Scenario and Scaling Diagnostics

Figure 7 shows that the difficulty ranking across scenarios is consistent at every tier: scarce (hardest) $>$ balanced $>$ rich (easiest). This validates the dataset design. Hybrid GA consistently dominates GA (original) in medium and large tiers across scenarios; small-tier outcomes are mixed.

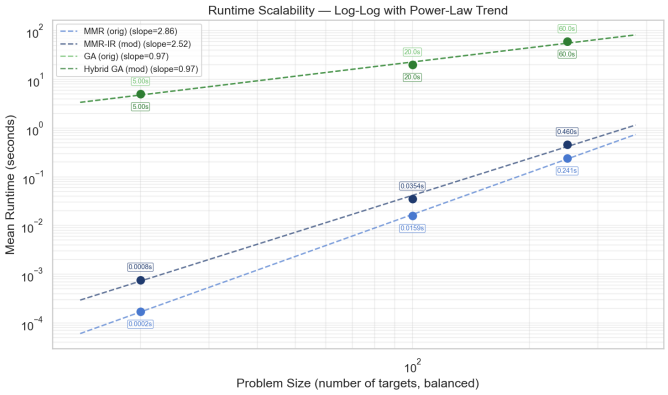


Fig. 6: Mean runtime (seconds) on a log-log scale. MMR and MMR-IR remain orders of magnitude faster than GA variants at all sizes. GA variants approach the time budget as size grows, while Hybrid GA still returns superior quality within the same budget.

Scaling Gain by Problem Size: Figure 8a and Figure 8b isolate the scaling behaviour of each modification. Hybrid GA's advantage grows sharply with problem size, confirming that the three modifications jointly address the scalability bottleneck of standard GA. MMR-IR shows a stable positive trend at every scale.

F. Why GA (Original) Underperforms at Scale

At large tier ($W = 375$, $T = 250$), the standard GA population size is $\max(375, 250) = 375$. Within a 60-second budget, this yields only $\approx$160 generations from a purely random start. Combined with the absence of elitism (the best solution can be overwritten) and uniform mutation (ignoring high-threat targets), the standard GA rarely converges near the MMR-greedy solution, let alone beyond it.

Hybrid GA resolves each of these issues directly: MMR seeding provides a strong individual from generation 0; elitism ensures this individual persists; stagnation injection prevents premature convergence; and threat-proportional mutation focuses every mutation step on the most impactful assignments.

G. Statistical Significance

Using the one-sided Wilcoxon signed-rank test ("original $>$ modified", $\alpha = 0.05$):

- **MMR-IR vs. MMR** is significant in all $9/9$ categories.
- **Hybrid GA vs. GA** is significant in $6/9$ categories.
- The three non-significant categories are all small-tier: small_balanced ($p = 0.9734$), small_scarce ($p = 0.9978$), and small_rich ($p = 0.1387$).

VIII. Conclusion

This paper addressed the Weapon-Target Assignment problem from an algorithm engineering perspective, identifying structural weaknesses in two well-studied baseline approaches — MMR and standard GA — and proposing targeted modifications that provably improve performance.

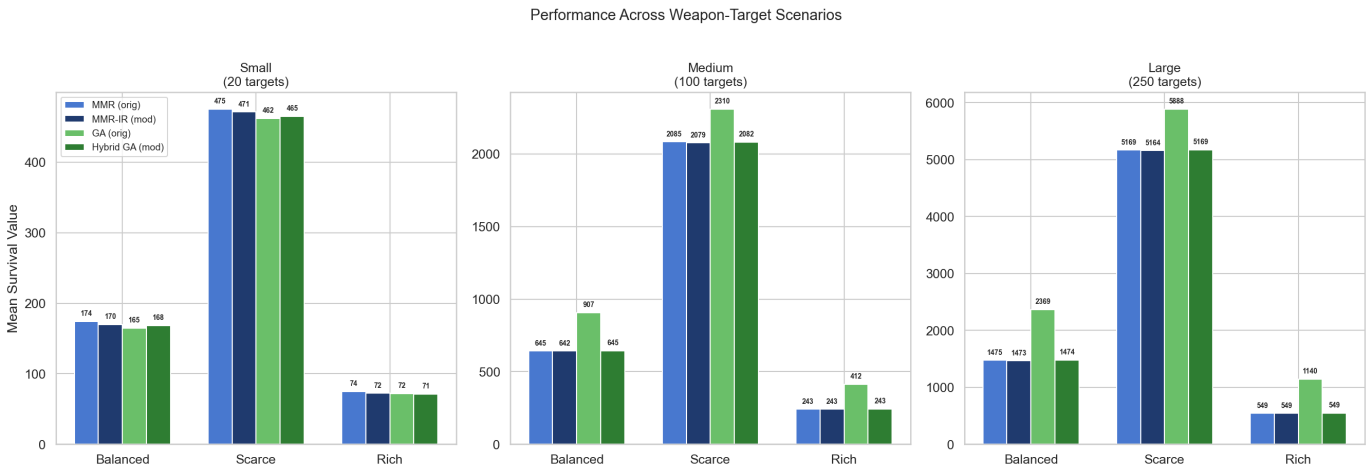
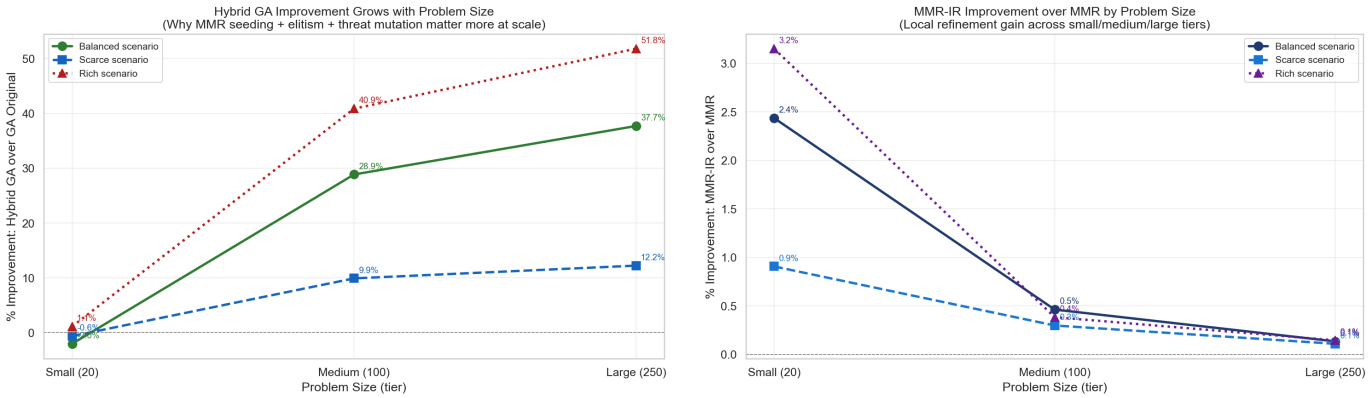


Fig. 7: Mean objective value broken down by scenario type (balanced, scarce, rich) and tier. The scarce scenario consistently produces the highest residual survival; the rich scenario the lowest. Algorithm rankings are stable across all conditions.



(a) Hybrid GA over GA.

(b) MMR-IR over MMR.

Fig. 8: Scaling comparison by problem size. Hybrid GA shows a strong scale-amplified gain; MMR-IR remains positive and steadier across scales.

MMR-IR augments the greedy heuristic with tie-aware allocation and two local-search phases (1-opt and 2-opt) using $O(1)$ incremental evaluation. On balanced tiers, it improves MMR by 2.44%, 0.46%, and 0.13% (small/medium/large), and remains statistically significant in all 9 categories.

Hybrid GA directly addresses the three failure modes of standard GA on large WTA instances – random initialisation, solution loss, and uninformed mutation – through MMR seeding, elitism with stagnation detection, and the novel threat-proportional mutation operator. On balanced tiers, it changes GA performance by -2.04%, 28.85%, and 37.70% (small/medium/large). The method is strongly effective in medium/large categories and significant in 6 of 9 categories overall.

Together, the two modifications improve the quality-runtime trade-off from different directions: MMR-IR provides consistent incremental gain at near-greedy speed, while Hybrid GA provides major gains where instance scale is large enough for guided evolutionary search to pay off.

Future directions. Promising extensions include coupling MMR-IR as a local-search operator within Hybrid GA (a memetic GA), exploring adaptive time-budget allocation that scales with instance hardness, and evaluating the modifications on dynamic WTA variants where targets arrive in real time.

Reproducibility

All code, datasets, and instructions are available at:

https://github.com/Arimotokika/CSE_462-Algorithm-Engineering-Sessional

All 180 instances, the four algorithm implementations, and the analysis pipeline are reproduced by executing the single master script:

```
python run_all.py
```

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